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Silvia Onorato

Curriculum Vitae

I am a post-doc at Gemini Observatory, mainly working with JWST/NIRSpec IFU data of high redshift quasars. In my Ph.D., I have developed skills in handling optical and near-infrared high-z quasar spectra, to constrain the fraction of neutral hydrogen, and quasar lifetimes. I also worked on stellar astrophysics, in particular on the study of multiple stellar populations through ultraviolet and optical photometry.

10 papers, of which 4 as a first-author; a complete publication list is enclosed and available on [NASA/ADS](https://www.nasa.gov/ads).

Work experience

- 2025–present **Postdoctoral researcher**, *Gemini Observatory/NSF NOIRLab*, Hilo, Hawaii, USA
Topic: High-redshift quasar studies with JWST; Supervisor: Dr. Emanuele Paolo Farina.
- 2022–present **Telescope observations**, *Leiden Observatory/Gemini Observatory*
We observed for two nights remotely with Keck/LRIS, and two nights on-site with Gemini/GNIRS, GMOS, IGRINS-2.
- 2018–present **Science outreach**, *University of Bologna/Gemini Observatory*
Speaker at guided tours in the Specola Astronomical Museum in Bologna, “Journey Through the Universe”, Gemini AstroDay Booth, and Astronomy on Tap in Hilo.
- 2022–2024 **Teaching assistant**, *Leiden Observatory*, Leiden, The Netherlands

Education

- 2021–2025 **Ph.D. in Astrophysics**, *Leiden Observatory*, Leiden, The Netherlands
Thesis: “*Chronicles of Cosmic Dawn - High redshift quasars as probes of supermassive black holes and the intergalactic medium*”; Supervisors: Prof. Joseph F. Hennawi, Prof. Joop Schaye.
- 2018–2021 **MSc in Astrophysics and Cosmology**, *Alma Mater Studiorum University of Bologna*, Bologna, Italy, Final grade: 109/110. GPA: 28.24/30.
Thesis: “*The primordial structure of Multiple Populations in the Globular Cluster NGC 2419*”; Supervisors: Dr. Mario Cadelano, Dr. Emanuele Dalessandro, Prof. Barbara Lanzoni
- 2015–2018 **BSc in Astronomy**, *Alma Mater Studiorum University of Bologna*, Bologna, Italy, Final grade: 106/110. GPA: 27.35/30.
Thesis: “*Mechanisms of Energy Transport in Astrophysics*”; Supervisor: Prof. Daniele Dallacasa

Scientific collaborations

- 2025–present **Member of the Aether collaboration**
- 2024–present **Member of the EREBUS collaboration**
- 2021–2025 **Member of the Pypelt collaboration**

Honours and grants

- 2026 **Travel funding for invited IASF Seminar**, Palermo, Italy
- 2026 **Travel funding for "Charting Cosmic Dawn in Copenhagen" conference**, Helsingør, Denmark
- 2024 **Travel funding for "Probing the Genesis of Supermassive Black Holes" conference**, Kavli IPMU, Kashiwa, Japan
- 2017 **Incentives to deserving students enrolled in the a.y. 2015/16 in courses belonging to classes L-41, L-27, L-30 and L-35**
 Winner of the scholarship: "Notice of competition for the allocation of incentives for deserving students enrolled in courses of study in the a.y. 2015/16 of special national and community interest at the University of Bologna" issued by Alma Mater Studiorum.

Telescope experience

I have reduced data obtained from the following facilities: HST WFC3/UVIS, VLT/X-Shooter, Gemini/GNIRS, Keck/NIRES, Gemini/GMOS, Keck/LRIS, Keck/DEIMOS, LBT/MODS, LBT/LBC, MPI/WFI, VLT/GIRAFFE, Chandra, VLA.

Digital skills

Analysis of optical and near-infrared spectra with PyPeIt.

Analysis of ultraviolet and optical data with DAOPHOT and IRAF.

Analysis of X-ray data with CIAO and XSPEC; radio data with AIPS.

Knowledge of Python, Fortran90, R, GitHub, L^AT_EX, and Microsoft Office.

Knowledge of DS9, Ginga, TopCat, Galfit, Sextractor, Sculptor, SuperMongo, and Cata-Pack.

Colloquia/Conference contributions

Invited talk

- 2026 **IASF Seminar**, IASF, Palermo, Italy
- 2026 **ITC Luncheon Talk**, Center for Astrophysics | Harvard-Smithsonian, Cambridge, MA
- 2026 **MAT Seminar**, MIT Kavli, Cambridge, MA
- 2024 **Colloquium**, University of Bologna/INAF-OAS, Bologna, Italy

Contributed talk

- 2026 **EREBUS Collaboration Meeting**, Ann Arbor, Michigan, USA
- 2026 **Supermassive Black Holes and Blue Notes**, Campus MIL, Université de Montréal, Montréal, Québec, Canada
- 2026 **Charting Cosmic Dawn in Copenhagen**, Helsingør, Denmark
- 2025 **HERA 25 - The Physics of Galaxies at the Epoch of Reionization**, Scuola Normale Superiore, Pisa, Italy
- 2025 **EREBUS/COSMOS-3D meeting**, INAF-OAS, Bologna, Italy
- 2024 **Probing the Genesis of Supermassive Black Holes: Emerging Perspectives from JWST and Expectation toward New Wide-Field Survey Observations**, Kavli IPMU, Kashiwa, Japan
- 2024 **The Origin and Evolution of Supermassive Black Holes**, Sexten, Italy
- 2024 **Massive Black Holes in the First Billion Years**, Kinsale, Ireland

- 2023 **NOVA Fall School**, *ASTRON*, Dwingeloo, The Netherlands
- 2023 **Young Astronomers on Galactic Nuclei**, *YAGN23*, Palermo, Italy
- 2023 **Reionisation in the Summer**, Heidelberg, Germany
- 2022 **Panchromatic view of the life-cycle of AGN**, *ESAC*, Madrid, Spain
- Lightning talk
- 2024 **Cosmic Dawn Revealed by JWST: The Physics of the First Stars, Galaxies, and Black Holes**, *KITP*, Santa Barbara, USA
- 2024 **European Astronomical Society Annual Meeting 2024**, *EAS24*, Padua, Italy
- 2023 **First Light in the Early Universe**, *MIT*, Cambridge, USA
- 2022 **What Drives the Growth of Black Holes?**, Reykjavík, Iceland

Proposals

- 2026 **Co-I of accepted Chandra Cycle 28 proposal**, *X-raying the cosmic dance of the highest redshift quasar pair known*
Proposal ID: 28700144; 01/01/2027 - 31/12/2027. PI: Fabio Vito
- 2025 **Co-I of accepted JWST Cycle 4 proposal**, *Ushering in the JWST Era of Precision Constraints on Reionization: A Survey of Quasar IGM Damping Wings at $6.5 < z < 7.4$*
Proposal ID: 9180; 12 months; 01/07/2025 - 30/06/2026. PI: Joseph F. Hennawi
- 2021 **Co-I of accepted DOLORES/LBT proposal**, *Does the host galaxy set the IMF of star clusters?*
Proposal ID: IT-2021B-039; 12h; 30/05/2022 - 03/09/2023. PI: Mario Cadelano

Other activities and Service

- 2026–present **Journal Club**, *Gemini Observatory*, Hilo, Hawaii, USA
Organizer of weekly journal club at Gemini Observatory.
- 2026–present **Reviewer for MNRAS**
- 2026–present **Paddling**, *Puna Canoe Club*, Hilo, Hawaii, USA
- 2022–2023 **Borrel Committee**, *Leiden Observatory*, Leiden, The Netherlands
Organizer of weekly social gatherings for colleagues. Responsible for providing food and drinks, scheduling activities, and inviting people to the events.
- 2004–2026 **Volleyball player**
2026: Hailo (Hilo). 2023–2025: SKC (Leiden). 2017: PGS Welcome (Bologna). 2004–2015: U.S. Volley (Bagheria).

Languages

- Italian Mother tongue
- English Fluent

Other interests

I am very passionate about volleyball, paddling, skiing, and cycling. While I discovered the first of my passions long ago, the last three proved to me that I can always learn new beautiful things, as I learned paddling in 2026, skiing in 2023 and cycling in 2022. From my Sicilian heritage, I inherited a deep connection with the sea, but recently I also discovered I like to hike to the mountains. I really love traveling and discovering new places to take pictures of. In my chilling time, I love to read books and watch TV series.

References

Emanuele Paolo Farina, *Gemini Observatory/NSF NOIRLab* emanuele.farina@noirlab.edu
Joseph F. Hennawi, *UCSB/Leiden Observatory* joe@physics.ucsb.edu
Mario Cadelano, *University of Bologna/INAF-OAS* mario.cadelano@unibo.it
Jan-Torge Schindler, *Hamburg Observatory* jtschindler@hs.uni-hamburg.de

Publications

Published/Submitted

1. **Silvia Onorato**, Mario Cadelano, Emanuele Dalessandro, Enrico Vesperini, Barbara Lanzoni, and Alessio Mucciarelli 2023. The structural properties of multiple populations in the dynamically young globular cluster NGC 2419. *Astronomy and Astrophysics*, 677, A8.
2. **Silvia Onorato**, Joseph F. Hennawi, Jan-Torge Schindler, Jinyi Yang, Feige Wang, Aaron J. Barth, et al. (21 authors) 2025. Optical and near-infrared spectroscopy of quasars at $z > 6.5$: public data release and composite spectrum. *Monthly Notices of the Royal Astronomical Society*, staf787.
3. **Silvia Onorato**, Joseph F. Hennawi, Elia Pizzati, Bram P. Venemans, and Anna-Christina Eilers 2026. Homogeneous measurements of proximity zone sizes for 59 quasars in the Epoch of Reionization. *Monthly Notices of the Royal Astronomical Society*, stag388.
4. **Silvia Onorato**, Emanuele Paolo Farina, Roberto Decarli, Klaudia Protušová, Elia Pizzati, Debora Pelliccia, et al. (35 authors) 2026. *Nāpōwawā'enakaulua*: a close quasar pair at cosmic dawn in the Aether survey. *ArXiv e-print*; accepted for publication in *ApJL*.
5. Suk Sien Tie, Joseph F. Hennawi, Feige Wang, **Silvia Onorato**, Jinyi Yang, Eduardo Bañados, Frederick B. Davies, Jose Oñorbe 2024. First measurement of the Mg II forest correlation function in the Epoch of Reionization. *Monthly Notices of the Royal Astronomical Society*, stae2193.
6. Silvia Belladitta, Eduardo Bañados, Zhang-Liang Xie, Roberto Decarli, **Silvia Onorato**, et al. (21 authors) 2025. Discovery and Characterization of 25 new Quasars at $4.6 < z < 6.9$ from Wide-Field Multi-Band Surveys. *Astronomy and Astrophysics*, 699, A335.
7. Jiamu Huang, Joseph F. Hennawi, Elia Pizzati, et al. (incl. **Silvia Onorato**, 24 authors) 2026. Clustering of $z \sim 6.6$ Quasars and [O III] Emitters Constrains Host Halo Masses and Duty Cycles in 25 ASPIRE Fields. *ArXiv e-print*; submitted on *MNRAS*.
8. Hyewon Suh, Roberto Decarli, Emanuele Paolo Farina, et al. (incl. **Silvia Onorato**, 18 authors) 2026. Rapid growth in a dual AGN during a gas-rich merger at $z \sim 4.5$. *ArXiv e-print*; accepted for publication in *Nature Astronomy*.
9. Aurora Mata-Sanchez, Emanuele Paolo Farina, Anniek J. Gloudemans, et al. (incl. **Silvia Onorato**, 20 authors) 2026. To be lensed or not to be lensed: on the nature of the alleged high- z lensed quasars J0109–5424 and P170+20. *ArXiv e-print*; accepted for publication in *ApJ*.
10. Anniek J. Gloudemans, Emanuele Paolo Farina, Roberto Decarli, et al. (incl. **Silvia Onorato**, 25 authors) 2026. Extreme Enrichment at Cosmic Dawn: Three Fell-Strong Quasars at $z > 6$ from the JWST Aether survey. *ArXiv e-print*; accepted for publication in *ApJ*.

In preparation

1. **Silvia Onorato**, Timo Kist, Joseph F. Hennawi, and Frederick B. Davies in prep. The Ly α transmission probability distribution function from the E-XQR-30 sample.
2. **Silvia Onorato**, Emanuele P. Farina, et al. in prep. *Nāpōwawā'enakaulua*: a multi-wavelength analysis.